

Free Body Diagram

- ☐ IDEALIZATIONS
- ☐ 2D SUPPORT REACTIONS
- ☐ FREE BODY DIAGRAM



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Free Body Diagram
Basic Mech. Engg.Idealizations
●●●2D Support Reactions
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Idealizations



PARTICLE

Idealizations are used in mechanics in order to simplify application of the theory

A particle has mass, but its size can be neglected



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Idealizations

RIGID BODY

Idealizations are used in mechanics in order to simplify application of the theory

A rigid body is a combination of a large number of particles in which all the particles remain at a fixed distance from one another, both before and after applying a load



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Idealizations

CONCENTRATED FORCE

Idealizations are used in mechanics in order to simplify application of the theory

A Concentrated Force is assumed to act at a point on a body



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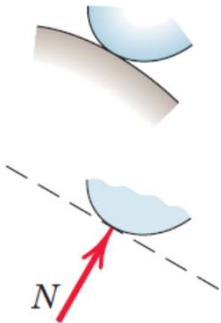
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SMOOTH SURFACES

Contact force is compressive and is normal to the surface



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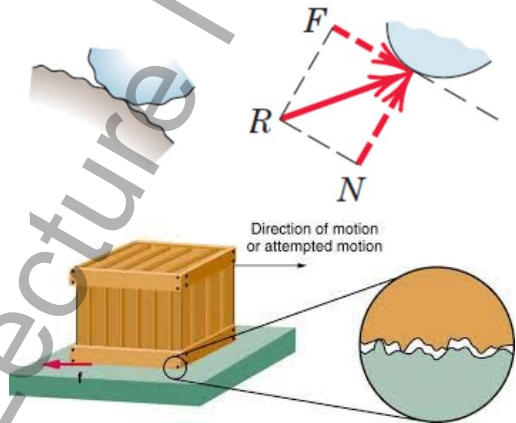
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ROUGH SURFACES

Rough surfaces are capable of supporting a tangential component F (frictional force) as well as a normal component N of the resultant contact force R .



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2D Support Reactions

ROLLER SUPPORT

Roller, rocker, or ball support transmits a compressive force normal to the supporting surface.





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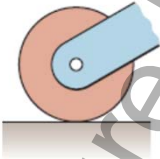
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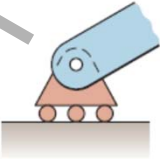
Free Body Diagram
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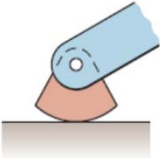
2D Support Reactions

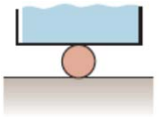
ROLLER SUPPORT

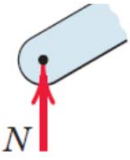
Roller, rocker, or ball support transmits a compressive force normal to the supporting surface.

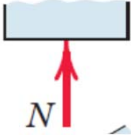












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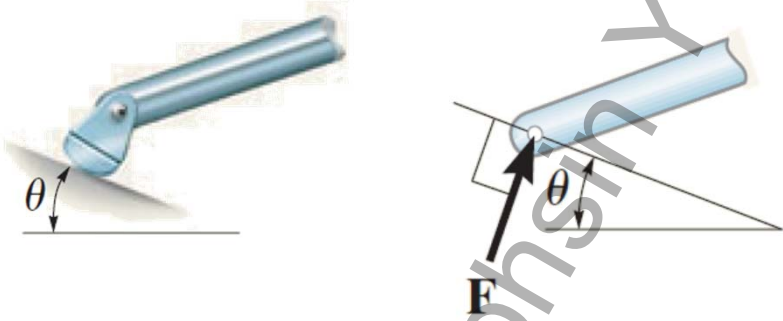
2D Support Reactions
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2D Support Reactions

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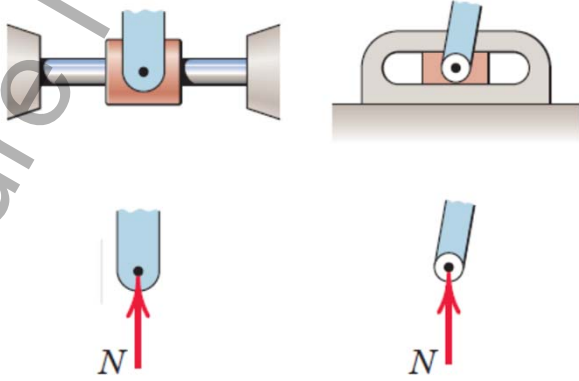
2D Support Reactions
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2D Support Reactions

FREELY SLIDING GUIDE

Collar or slider free to move along smooth guides; can support force normal to guide only.



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2D Support Reactions

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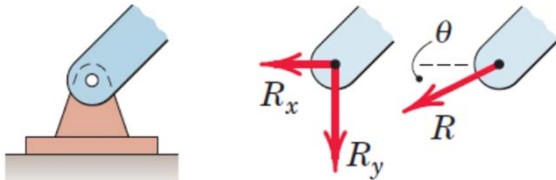
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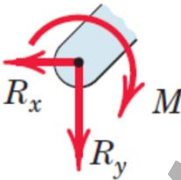


PIN CONNECTION

A freely hinged pin connection is capable of supporting a force in any direction in the plane normal to the pin axis.



A pin not free to turn also supports a couple M.



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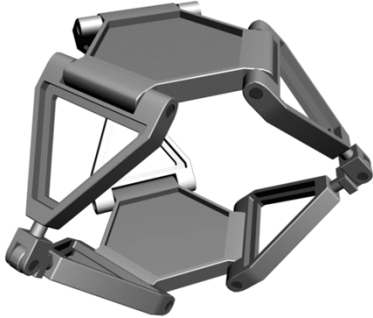
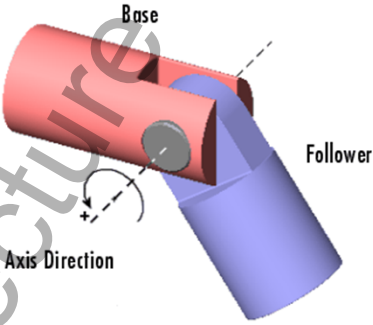
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2D Support Reactions



PIN CONNECTION

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2D Support Reactions

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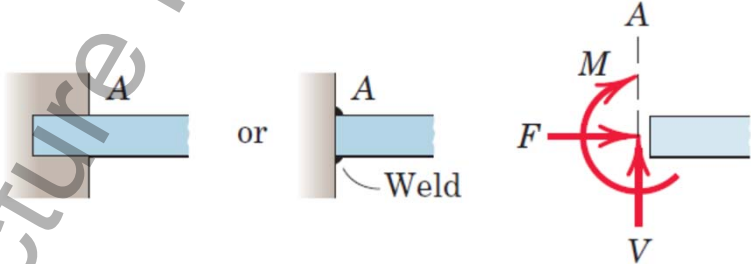
2D Support Reactions
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2D Support Reactions

BUILT-IN OR FIXED SUPPORT

A built-in or fixed support is capable of supporting an axial force F , a transverse force V (shear force), and a couple M (bending moment) to prevent rotation



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2D Support Reactions

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2D Support Reactions

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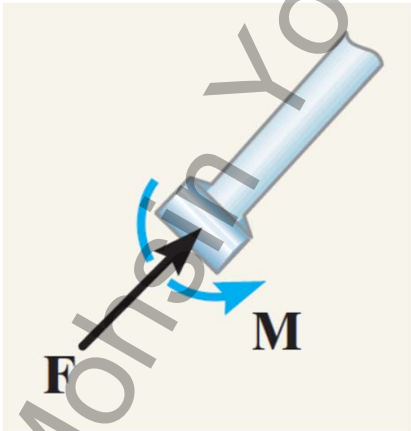
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2D Support Reactions

MEMBER FIXED CONNECTED TO COLLAR

The reactions are the couple moment and the force which acts perpendicular to the rod.



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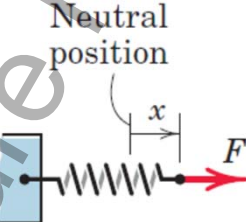
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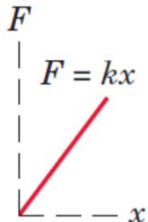
2D Support Reactions

SPRING ACTION

For a linearly elastic spring the stiffness k is the force required to deform the spring a unit distance.

Neutral position





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2D Support Reactions

FLEXIBLE CABLE, BELT, CHAIN, OR ROPE

Force exerted by a flexible cable is always a tension away from the body in the direction of the cable.



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GRAVITATIONAL ATTRACTION

The resultant of gravitational attraction on all elements of a body of mass m is the weight $W = mg$ and acts toward the center of the earth through the center mass G .



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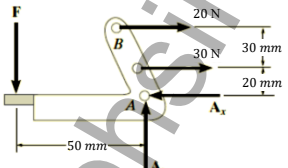
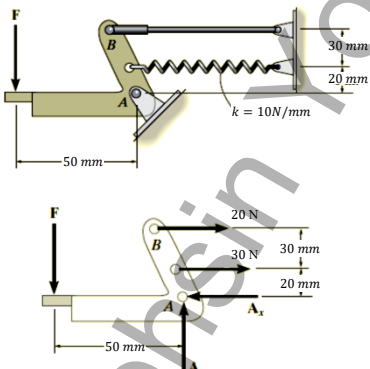
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Free Body Diagram

INTRODUCTION

Specification of all the known and unknown external forces that act on the body



A thorough understanding of how to draw a free-body diagram is of primary importance for solving problems in mechanics

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Free Body Diagram

PROCEDURE

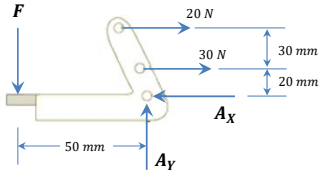
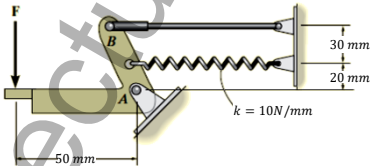
1 – Establish Coordinate Axis

2 – Isolate Body & Draw Outlined Shape

3 – Draw All Applied Forces and Moments

4 – Identify Support Reactions

5 – Indicate Dimensions Necessary for Computing



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Free Body Diagram

EXAMPLE I

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Free Body Diagram

EXAMPLE II

Draw the free-body diagram of the uniform beam. The beam has a mass of 100 kg.

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Free Body Diagram

EXAMPLE II

Draw the free-body diagram of the uniform beam. The beam has a mass of 100 kg.

Diagram of a beam of length 6 m. A downward force of 1200 N is applied at a distance of 2 m from support A. The weight of the beam (981 N) acts at the center (3 m from A).

Free-body diagram of the beam. The beam is represented by a horizontal line. At support A (left end), there are reaction forces A_x (horizontal, pointing right) and A_y (vertical, pointing up), and a reaction moment M_A (curved arrow, counter-clockwise). At a distance of 2 m from A, there is a downward force of 1200 N. At a distance of 3 m from A (the center of the beam), there is a downward force of 981 N. The beam is labeled with 'x' and 'y' axes at support A.

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Free Body Diagram

EXAMPLE III

Two smooth pipes, each having a mass of 300 kg, are supported by the forked tines of the tractor in Fig. 5–9a. Draw the free-body diagrams for each pipe and both pipes together.

Photograph of two large pipes being lifted by a tractor's forked tines.

Diagram of two pipes, A and B, each with a radius of 0.35 m. They are supported by a forked tine that is inclined at an angle of 30° to the horizontal.

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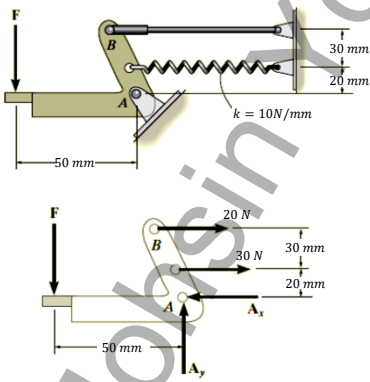
Free Body Diagram
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EXAMPLE IV

Draw the free-body diagram of the foot lever. The operator applies a vertical force to the pedal so that the spring is stretched 3 mm. and the force in the short link at B is 20 N.



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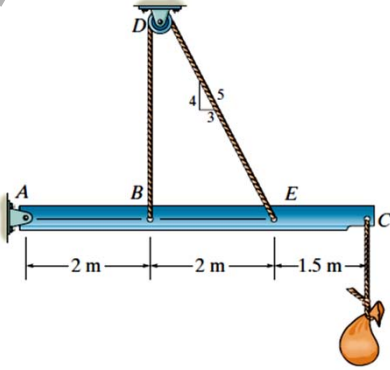
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EXAMPLE V

Draw the free-body diagram of the beam which supports the 80-kg load and is supported by the pin at A and a cable which wraps around the pulley at D.



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Free Body Diagram

PRACTICE PROBLEMS

Problems 5.1, 5.3 to 5.5, 5.8

Prob. 5-1

Prob. 5-4

Prob. 5-8

Prob. 5-3

Prob. 5-5

Russell C. Hibbeler. Engineering mechanics: Statics and dynamics 12th edition, pp 211

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MORE PRACTICE PROBLEMS

Determine the support reactions at the fixed support, A . Neglect the weight and size of the beam.

200 N·m

80 N/m

1000 N

2 m

1.5 m

4 m

$A_x = 600\text{ N}, A_y = 920\text{ N}, M_A = 3730\text{ Nm}$

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Mohsin Yousuf

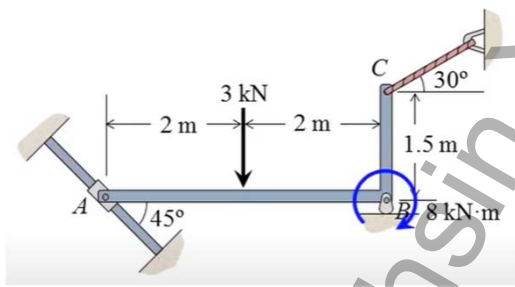
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Free Body Diagram

MORE PRACTICE PROBLEMS

Q. Determine the support reactions at the smooth collar at A , rocker at B , and cable at C . Neglect the weight and size of the member ABC .



$F_A = -1.13 \text{ kN}, F_B = 3.34 \text{ kN}, T_C = 0.924 \text{ kN}$

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